

## Python Programming Assignment - 25

<1> what is object-oriented programming?

→ Python is object-oriented language, allowing to structure the code, maintain the code and allows to build reusable applications with less code.

<2> what are the four core characteristics of OOP?

→ Python is multi-Paradigm programming. it's allow to write a code in four way

- (a) Procedural Programming
- (b) Functional Programming
- (c) Object Oriented Programming
- (d) Scripting Programming.

<3> Explain class & object with real world analogy?

→ Class is as blueprinted map or stencil, and ~~an~~ Object as the real, physical item you build using that blueprint.  
example: The blueprint is just paper & lines you can't sleep inside, cook or open a window. and now a construction company uses that blueprint to build real houses the build house No. 1, 2, 3

<4> what is constructor in python? why is `__init__()` used?

→ `__init__` is reserved method in python classes. It is known as a constructor in object oriented concepts.

The method called when an object is created from the class and it allows the class to initialise the attributes of a class.

<5> Is constructor mandatory in a class? Explain?

→ No, constructor <sup>is NOT</sup> mandatory in a class. If you do not explicitly write an `__init__` method, Python automatically creates a hidden empty default constructor behind the scenes so you can still create objects without crashing.

<6> What is a destructor? When is `__del__()` called?

→ A destructor is a special method in object-oriented programming that handles clean-up operations when an object is destroyed or removed from memory.

Python's "garbage collector" will delete objects to free up computer's RAM, and trigger the `__del__()` method.

<7> Difference between constructor & destructor?

→ The core difference is that a constructor handles the initialization of an object, while a destructor handles the clean-up of the object when it is removed from memory.

<8> what are instance variables and class variables?

- The core difference instance variable belongs to a object
- <1> while class variable are shared by all objects created from the class.

<9> Explain instance methods, class methods & static method?

<:-> Instance method: ~~requires~~ receives self as a first argument to access and modify object-specific state.

class method: receives cls as an first argument and manage level state.

static method: static method can access both instance and class variable.

<10> why do we use @classmethod and @staticmethod decorators?

- Both decorators we use for to change how python passes arguments to methods, allowing us to build cleaner architecture, save memory and create alternative way to initialize objects.